

Differentiation

Week 3B

Overview

Techniques of Derivatives

Key Ideas - Basic Applications of Derivatives

- Higher Order Derivatives

- Rule 3: Product Rule

Key Idea – Basic Applications of Derivatives of Powers of x

Now that we are familiar with using the power rule to differentiate, we can apply this rule to find the gradient of a curve at a specific point or to find the x -value when the gradient is given.

Discussion Question

For the function $f(x) = \frac{x^3}{2}$ evaluate $f(2)$ and $f'(2)$ and explain the difference between what is being calculated in each instance.

$$f(2) = \frac{2^3}{2} = 4$$

$f(2)$ gives the output value of the function when $x = 2$

$$f'(x) = \frac{3x^2}{2}$$

$$f'(2) = \frac{3 \times 2^2}{2} = 6$$

$f'(2)$ gives the gradient of the function at $x = 2$

For the function $f(x) = \frac{x^3}{2}$ evaluate x when $f(x) = 2$ and $f'(x) = 2$ and explain the difference between what is being calculated in each instance.

$$f(x) = 2 \Rightarrow \frac{x^3}{2} = 2$$

$$\Rightarrow x^3 = 4 \\ x = \sqrt[3]{4}$$

$f(x) = 2$ is finding the value of x that makes the function equal to 2

OR what input gives an output of 2

$$f'(x) = \frac{3x^2}{2}$$

$$f'(x) = 2 \Rightarrow \frac{3x^2}{2} = 2$$

$$\Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \pm \sqrt{\frac{4}{3}}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

$f'(x) = 2$ means the tangent to the function has a gradient of 2 at $x = \pm \frac{2}{\sqrt{3}}$

Examples: Find the derivative of $f(x) = x^3 + 3x$ at the point when $x = 2$.

$$f'(x) = 3x^2 + 3$$

Now substitute $x = 2$ into $f'(x) = 2x + 6$

$$\begin{aligned}\text{ie. } f'(2) &= 3 \times 2^2 + 3 \\ &= 15\end{aligned}$$

Examples: Find $f'(4) = \frac{\sqrt{x}}{2}$.

$$f(x) = \frac{\sqrt{x}}{2} = \frac{1}{2}x^{1/2}$$

$$\begin{aligned}f'(x) &= \frac{1}{2} \times \frac{1}{2}x^{-1/2} \\ &= \frac{1}{4\sqrt{x}}\end{aligned}$$

Now substitute $x = 4$ into $f'(x) = \frac{1}{4\sqrt{x}}$

$$\text{ie. } f'(4) = \frac{1}{4\sqrt{x}} = \frac{1}{4\sqrt{4}} = \frac{1}{8}$$

Examples: If $f(x) = (x + 3)^2$, find the value of x for which the gradient is zero, i.e $f'(x) = 0$.

$$\begin{aligned}f(x) &= (x + 3)^2 \\&= x^2 + 6x + 9 \\f'(x) &= 2x + 6\end{aligned}$$

Now solve $f'(x) = 0$ ie. $2x + 6 = 0$

$$2x = -6$$

$$x = -3$$

Examples: Find the coordinates of the point on the curve $y = x^2 - 1$ where the gradient is equal to -8 .

$$\begin{aligned}y &= x^2 - 1 \\y' &= 2x\end{aligned}$$

Now solve $y' = -8$ ie. $2x = -8$

$$x = -4$$

Now substitute $x = -4$ into the original function.

$$y = x^2 - 1 = (-4)^2 - 1 = 15$$

Therefore, the coordinates are $(-4, 15)$

Example: At what points does the curve $y = 2x^2 - 8$ cut the x -axis? Find the gradient of the curve at each of these points.

The curve cuts the x -axis at $y = 0$.

$$2x^2 - 8 = 0$$

$$2x^2 = 8$$

$$x^2 = 4 \rightarrow x = \pm 2$$

To find the gradient of the curve we need to differentiate $y =$

$$y' = 4x$$

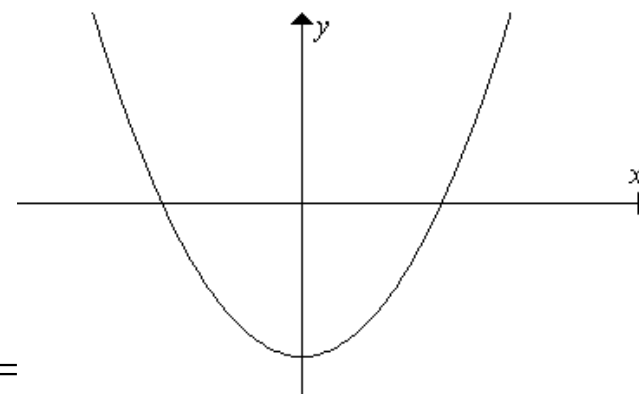
Now substitute $x = \pm 2$ into $y' = 4x$.

$$\begin{aligned} y'(2) &= 4 \times 2 \\ &= 8 \end{aligned}$$

$$\begin{aligned} y'(-2) &= 4 \times -2 \\ &= -8 \end{aligned}$$

Gradient at $x = 2$ is 8

Gradient at $x = -2$ is -8



Exercise:

- (a) Find the gradient of the tangent to the curve $y = \frac{4}{x^2} - x - 8$ at the point where $x = -1$.

$$y = \frac{4}{x^2} - x - 8 \Rightarrow y = 4x^{-2} - x - 8$$

$$y' = -8x^{-3} - 1$$

$$= \frac{-8}{x^3} - 1$$

$$\text{at } x = -1 \quad y' = \frac{-8}{(-1)^3} - 1$$

$$= 1$$

- (b) Determine $y'(1)$ if $y = x\sqrt{x}$.

$$y = x^1 \times x^{\frac{1}{2}}$$

$$y = x^{1+\frac{1}{2}}$$

$$y = x^{\frac{3}{2}}$$

$$y = x^{\frac{3}{2}}$$

$$y' = \frac{3}{2} x^{\frac{1}{2}}$$

$$y'(1) = \frac{3}{2} \times 1^{\frac{1}{2}}$$

$$= \frac{3}{2}$$

Exercise:

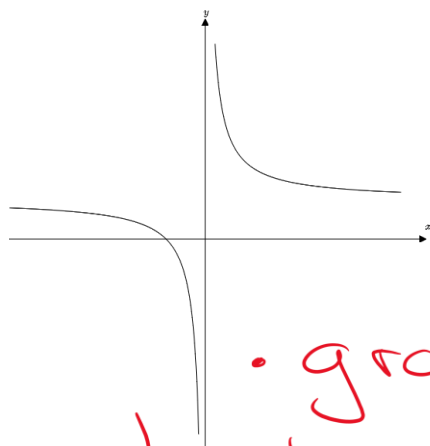
- (a) If $y = 3x^{-2} - 5$, find the value of x when $\frac{dy}{dx} = \frac{3}{4}$

$$\begin{aligned}\frac{dy}{dx} &= 3 \times (-2)x^{-3} \\ &= -6x^{-3} \\ &= \frac{-6}{x^3}\end{aligned}$$

$$\begin{aligned}\text{now } \frac{-6}{x^3} &= \frac{3}{4} \\ \Rightarrow 3x^3 &= -24\end{aligned}$$

$$\begin{aligned}x^3 &= -8 \\ x &= \sqrt[3]{-8} \\ x &= -2\end{aligned}$$

- (b) Find the gradient of the curve $y = \frac{1}{x} + 1$ at the point where the curve crosses the x -axis.



• "crosses the x -axis" is the same as finding the x -intercept i.e. let $y = 0$

$$y = \frac{1}{x} + 1 = 0 \Rightarrow \frac{1}{x} = -1$$

• gradient

$$\begin{aligned}y &= \frac{1}{x} + 1 = x^{-1} + 1 \\ y' &= -x^{-2} = \frac{-1}{x^2}\end{aligned}$$

$$\begin{aligned}\text{at } x &= -1 \quad \left| \begin{array}{l} 1 = -x \\ x = -1 \end{array} \right. \\ y' &= \frac{-1}{(-1)^2} \\ &= -1\end{aligned}$$

Exercise:

Find the coordinates of the point on the parabola $y = x^2 - 2x - 8$ at which the tangent is parallel to the line $2x + y = 7$.

- $y = x^2 - 2x - 8 \Rightarrow y' = 2x - 2$
- find the slope of $2x + y = 7$:
make y the subject $y = -2x + 7$
 $\therefore m = -2$
- now solve $\underline{2x - 2 = -2}$ as tangent is parallel to $2x + y$ i.e. $y' = -2$
$$2x - 2 = -2$$
$$x = 0$$
- find y -coord of $x = 0$:
$$y = 0^2 - 2 \times 0 - 8$$
$$= -8$$

 $\therefore (0, -8)$ is the coordinate

Key Idea – Higher Order Derivative

If $f(x)$ is a differentiable function, then effectively we can differentiate it many times!

Derivatives and common notations used:

Notation:

For $y = f(x)$,

First Derivative: $f'(x) = y' = y^{(1)} = f^{(1)}(x) = \frac{dy}{dx}$

Second Derivative: $f''(x) = y'' = y^{(2)} = f^{(2)}(x) = \frac{d^2 y}{dx^2}$

Third Derivative: $f'''(x) = y''' = y^{(3)} = f^{(3)}(x) = \frac{d^3 y}{dx^3}$

Beyond about fourth derivative, the use of prime superscripts (as in f' , f'' , f''' and f'''') becomes increasingly awkward and we switch to the use of superscripted numeral in parentheses (as $f^{(4)}$ and $f^{(5)}$ and so on).

Fourth Derivative: $f^{(4)}(x) = y'''' = y^{(4)} = f^{(4)}(x) = \frac{d^4 y}{dx^4}$

n th Derivative : $f^{(n)}(x) = y^{(n)} = \frac{d^n y}{dx^n}$

Example:

Find the 1st, 2nd, 3rd and 4th derivative of $y = x^4 + x^2 - 4x + 1$.

$$f'(x) = 4x^3 + 2x - 4$$

$$f''(x) = 12x^2 + 2$$

$$f'''(x) = 24x$$

$$f^{(4)}(x) = 24$$

It is clear that we will have $f^{(n)}(x) = 0$ for all $n \geq 5$

Note: Once a derivative is zero, all higher derivatives are also zero!!!

Example:

(a) Given $y = 27x - x^3$, find $\frac{d^2y}{dx^2}$ at $x = 2$.

$$\frac{d^2y}{dx^2} = -6x$$

$$\text{at } x = 2, \frac{d^2y}{dx^2} = -6(2) = -12$$

(b) Given $y = x^3 + 1$, find the value of $y(1) - y''(1)$.

$$y = x^3 + 1$$

$$y' = 3x^2, \quad y'' = 6x$$

$$y(1) - y''(1)$$

$$= (1)^3 + 1 - 6(1)$$

$$= 1 + 1 - 6 = -4$$

Exercise:

- (a) Given $y = 5x^3 + 2x$ find $y^{(23)}(x)$.

$$\begin{aligned} y^{(1)} &= 15x^2 + 2 \\ y^{(2)} &= 30x \\ y^{(3)} &= 30 \\ y^{(4)} &= 0 \\ y^{(5)} &= 0 \\ &\vdots \\ y^{(23)} &= 0 \end{aligned}$$

- (b) Given $f(x) = \sqrt{x}$ find $f''(x)$ write your answer in surd form.

$$\begin{aligned} f(x) &= \sqrt{x} = x^{\frac{1}{2}} \\ f'(x) &= \frac{1}{2} x^{-\frac{1}{2}} \\ f''(x) &= \frac{1}{2} \times \left(-\frac{1}{2}\right) x^{-\frac{3}{2}} \\ &= -\frac{1}{4} x^{-\frac{3}{2}} \\ &= -\frac{1}{4x^{\frac{3}{2}}} = -\frac{1}{4\sqrt{x^3}} \end{aligned}$$

(c) Given $f(x) = 2x^4 + x^2 + x + 1$, evaluate $f'(2) - f''(1)$.

$$f'(x) = 8x^3 + 2x + 1$$

$$f'(2) = 8 \times 2^3 + 2 \times 2 + 1 = 69$$

$$f''(x) = 24x^2 + 2$$

$$f''(1) = 24 \times 1^2 + 2 = 26$$

$$\begin{aligned} \therefore f'(2) - f''(1) &= 69 - 26 \\ &= 43 \end{aligned}$$

Key Idea – Rule 3: Product Rule

We have already looked at two rules that apply to all differentiable functions. In this lecture, we will introduce three more rules.

We will use power functions to demonstrate these rules. However, it is important to remember that they apply to ALL differentiable functions.

If $u = f(x)$ and $v = g(x)$ are differentiable functions, then $u(x) \times v(x)$ can be differentiated using the **product rule**:

$$\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx} = f'(x)g(x) + f(x)g'(x)$$

In simplified form we could write

$$\frac{d}{dx}(uv) = u'v + uv'$$

Suggestion: Learn the product rule independently of the variable:

“Derivative of the first times the second plus the first times the derivative of the second.”

Example:

Differentiate $f(x) = (2x - 5)(3x + 2)$ using the product rule.

$$u = 2x - 5 \qquad v = 3x + 2$$

$$u' = 2 \qquad v' = 3$$

$$\begin{aligned} f'(x) &= u'v + uv' \\ &= 2 \times (3x + 2) + (2x - 5) \times 3 \\ &= 6x + 4 + 6x - 15 \\ &= 12x - 11 \end{aligned}$$

Note: This product can also be differentiated by firstly expanding it.

Exercise:

Differentiate $y = (5x^2 + 8)(x^2 - 3)$ using the product rule.

$u \times v$

$$u = 5x^2 + 8$$

$$v = x^2 - 3$$

$$u' = 10x$$

$$v' = 2x$$

product rule: if $y = u \times v$ $\frac{dy}{dx} = u'v + uv'$

$$\begin{aligned}\frac{dy}{dx} &= u'v + uv' \\ &= 10x \times (x^2 - 3) + (5x^2 + 8) \times 2x \\ &= 10x^3 - 30x + 10x^3 + 16x \\ &= 20x^3 - 14x\end{aligned}$$

Exercise:

Use the product rule to find the gradient of the tangent to the curve $f(x) = \sqrt{x}(x^2 - 2)$ at $x = 2$.

$u \times v$

$$f(x) = x^{\frac{1}{2}}(x^2 - 2)$$

$$u = x^{\frac{1}{2}}$$

$$v = x^2 - 2$$

$$u' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}} \quad v' = 2x$$

this is the same
as $f'(2)$

$$f'(x) = u'v + uv'$$
$$= \frac{1}{2\sqrt{x}}(x^2 - 2) + \sqrt{x}(2x)$$

$$f'(2) = \frac{1}{2\sqrt{2}}(2^2 - 2) + \sqrt{2}(2 \times 2)$$
$$= \frac{1}{2\sqrt{2}} \times 2 + \sqrt{2} \times 4 = \frac{1}{\sqrt{2}} + 4\sqrt{2}$$